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A new type of integral transform and its applications to the solution of differential equations

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Abstract

In this paper, we introduce a novel integral transform to address certain ordinary differential equations, and it has been demonstrated that this transform possesses unique and highly advantageous properties. We examine this fascinating integral transform and its effectiveness in solving linear ordinary and partial differential equations. Also, we provide a graphical depiction of the solution to the respective differential equations.

Keywords: Integral transform; Ordinary differential equations; Partial differential equations

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1 Introduction

There are numerous natural phenomena in our daily lives that are related to physics, engineering, astronomy, and the spread of epidemics, among other fields. Researchers have expressed interest in using mathematical models to describe these natural phenomena, as seen in references [1–6]. In order to understand and explain these phenomena, the mathematical models must be solved using various numerical methods, including integral transformations such as the Laplace transformation, we refer the readers to these articles [7–11]. The integral transform method is a commonly employed technique for solving differential equations with initial or boundary conditions, frequently represented as integral equations. Integral transforms are utilized to solve differential equations in various applications including control theory, system engineering and applied physics [12–15]. The concept of integral comes back to the work of P. S. Laplace [16] in 1780, where it was originally defined for a function Φ as follows:

$$\mathcal{L}(\phi(z)) = \Phi(p) = \int_0^{\infty} e^{-pz} \phi(z) dz, \quad (1.1)$$

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where $p \in \mathbb{C}$, and $z \geq 0$. Fourier in [17] 1822 is defined as

$$\mathcal{F}(\phi(z)) = \Phi(\zeta) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-i\zeta z} \phi(z) dz. \quad (1.2)$$

In recent years, many mathematicians are working on new transforms such as Sumudu, Atangana Kilicman, Natural, Shuehu transform, etc; see these works [12, 18–30], for more details. Maitama and Zhao [29] proposed a novel transform, referred to as the Shehu transform, which is defined for all functions $\phi(z)$ within a certain class, where a set $\mathcal{A}$ defined as

$$\mathcal{A} = \left\{ \phi(z) : \exists \mathcal{N}, \vartheta_1, \vartheta_2 > 0, |\phi(z)| < \mathcal{N} e^{\frac{|z|}{\vartheta_i}}, \text{ if } z \in (-1)^i \times [0, +\infty) \right\},$$

and $\mathcal{S}(\phi(z))$ and is given by

$$\mathcal{S}(\phi(z)) = \Phi(s, u) = \int_0^{+\infty} e^{-\frac{sz}{u}} \phi(z) dz.$$

In 2008, authors [21], introduced the natural transform, such that this transform is defined for $\phi(z)$ and given by the integral

$$\mathcal{NT}(\phi(z)) = \Phi(s, v) = \int_{-\infty}^{+\infty} e^{-sz} \phi(vz) dz,$$

where s, v are two variables and z is the frequency variable. Recently, Jadhav et al. obtained solutions of fractional differential equations in electrical and mechanical systems by using the Elzaki, Sumudu, and Laplace transform, see [31–33].

Several recent studies have highlighted the effectiveness of such approaches. Higazy et al. [1] applied the Sawi decomposition method to Volterra integral equations, demonstrating that hybrid analytical–numerical techniques can yield accurate and efficient solutions. Building on this, Jafari et al. [2] proposed the Upadhyaya integral transform, which proved to be a powerful tool for addressing nonlinear Volterra equations. More recently, in [34], the Anuj integral transform was introduced, demonstrating its simplicity and effectiveness in solving Abel's integral equation in classical mechanics. Aggarwal and Bhatnagar [35] explored dualities between the Laplace transform and several modern transforms, such as Sumudu, Aboodh, Mohand, and Sawi, highlighting their close interrelationships and practical relevance. Higazy et al. applied the Shehu transform to a model of HIV-1 infection and showed that it can effectively determine both the number of infected cells and the concentration of viral particles in plasma, obtaining exact solutions with reduced computational complexity, see [5]. Upadhyaya [36] introduced the Upadhyaya transform, which is a generalized form of the Laplace transform and unifies many existing variants, such as Shehu, Sawi, and Elzaki, offering greater flexibility and broader applicability in solving integral equations. Higazy and Aggarwal used the Sawi transform to solve systems of ordinary differential equations (SODEs), including a chemical reaction model, demonstrating that the Sawi transform can produce exact solutions of SODEs without large computational effort, see [37].

This paper gives a new integral transform of exponential order and some applications for solving differential equations. Our work is a generalization of the Shehu transform and natural transform [21, 29]. However, there are cases where the Laplace transform and the Elzacki transform can not solve integral equations, but they can be solved using this transformation. Therefore, it can be used as an effective tool for solving integro-differential equations. On the other hand, this new transformation also builds on the previous transformations, but on different domains. This transformation takes a different form when transforming an equation, obtaining the solution to the equation.

This work is organized into three main sections: The first section provides a general introduction. The second section introduces the definition of the new integral transform, along with its key properties and related results. Lastly, the third section illustrates how this transform is applied to solving differential equations.

2 Main results

In this section, we would declare that our analysis is within the set $\mathcal{F}$ which is defined as follows:

$$\mathcal{F} = \left\{ f(z) : \exists \mathcal{N}, \kappa_1, \kappa_2 > 0, |f(z)| > \mathcal{N} \exp\left(\frac{|z|}{\kappa_j}\right), \right. \\ \left. \text{if } z \in (-1)^{j-1} \times]-\infty, 0], j = 1, 2 \right\}, \quad (2.1)$$

where a real constant $\mathcal{N}$ must be a finite number, while κ_1, κ_2 may be finite or infinite.

Definition 1 Let the new integral transform denotes by the notation $\xi(\cdot)$, and a function f of exponential order is defined on a set $\mathcal{F}$. Then, the new transform of f is given by the integral equation:

$$\xi[f(z)](s, u) = \Theta(s, u) = \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) f(z) dz, \quad z \leq 0, \quad \frac{1}{\kappa_1} \leq u \leq \frac{1}{\kappa_2}, \quad (2.2)$$

provided that the limit of the integral exists, this means it converges; otherwise, it diverges.

The inverse of the novel transform (2.2) is defined as:

$$\xi^{-1}[\Theta(s, u)] = f(z). \quad (2.3)$$

Theorem 2.1 (*Existence of the New Transform*). Let $u \neq 0$, be fixed, if a function $f :]-\infty, 0] \rightarrow \mathbb{R}$ is piecewise continuous and is of exponential order α , i.e., there exist the constants $M > 0$, and $\alpha \in \mathbb{R}$, such that $|f(z)| \leq Me^{-\alpha z}$, for $z \leq 0$. Then, the new integral transform of f always exists for $u \neq 0$, and $\text{Re}\left(\frac{s}{u}\right) > \alpha$.

Proof To prove this result, we consider $\theta \leq z \leq 0$, so one has

$$\begin{aligned} |\xi[f(z)]| &= \left| \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) f(z) dz \right|, \quad z \leq 0 \\ &= \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) |f(z)| dz \end{aligned}$$

$$\begin{aligned}
&= \frac{M}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) \exp(-\alpha z) dz \\
&= \frac{M}{u^2} \int_{-\infty}^0 \exp\left(\frac{(s-\alpha u)z}{u}\right) dz \\
&= \frac{M}{u(s-\alpha u)} \lim_{\theta \rightarrow -\infty} \exp\left(\frac{(s-\alpha u)z}{u}\right) \Big|_{\theta}^0 \\
&= \frac{M}{u(s-\alpha u)}, \quad u \neq 0, \text{ and } \operatorname{Re}\left(\frac{s}{u}\right) > \alpha.
\end{aligned}$$

This completes the proof. $\square$

Theorem 1 Let $\xi[f(z)]$ be the new integral transform, then

1. $\xi[1] = \frac{1}{su}, \frac{s}{u} > 0;$
2. $\xi[z] = \frac{-1}{s^2};$
3. $\xi[z^n] = \frac{(-1)^n n! u^{n-1}}{s^{n+1}}, n \in \mathbb{N}.$

Proof By applying (2.2), we get

$$\begin{aligned}
1. \quad \xi[1] &= \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) dz = \left[\frac{1}{su} \exp\left(\frac{sz}{u}\right) \right]_{-\infty}^0 = \frac{1}{su}. \\
2. \quad \xi[z] &= \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) z dz = \frac{1}{s^2} \left[\left(\frac{sz}{u} - 1\right) \exp\left(\frac{sz}{u}\right) \right]_{-\infty}^0 = \frac{-1}{s^2}. \\
3. \quad \xi[z^n] &= \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) z^n dz = \frac{1}{su} \left[z^n \exp\left(\frac{sz}{u}\right) \right]_{-\infty}^0 - \frac{n}{su} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) z^{n-1} dz \\
&= \frac{n}{s^2} \left[z^{n-1} \exp\left(\frac{sz}{u}\right) \right]_{-\infty}^0 - \frac{n(n-1)}{s^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) z^{n-2} dz \\
&\vdots \\
&= \frac{(-1)^n n! u^{n-1}}{s^{n+1}}. \quad \square
\end{aligned}$$

Theorem 2 The new integral transform of an exponential function $\exp(az)$ is given by:

$$\xi[\exp(az)] = \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) \exp(az) dz = \frac{1}{u(s+au)}.$$

Theorem 3 The new integral transform of the trigonometric functions, such as $\cos(az)$ and $\sin(az)$ are defined by

$$\begin{aligned}
\xi[\cos(az)] &= \frac{s}{u(s^2 + a^2 u^2)}, \quad \text{and} \\
\xi[\sin(az)] &= \frac{-a}{(s^2 + a^2 u^2)},
\end{aligned}$$

with $\frac{s}{u} + a > 0$. In the case $a = 1$, we have

$$\begin{aligned}\xi[\cos(z)] &= \frac{s}{u(s^2 + u^2)}, \text{ and} \\ \xi[\sin(z)] &= \frac{-1}{(s^2 + u^2)}.\end{aligned}$$

Proof By definition (2.2), we have

$\xi[\cos(az)] = \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sZ}{u}\right) \cos(az) dz$, we pose $I = \int_{-\infty}^0 \exp\left(\frac{sZ}{u}\right) \cos(az) dz$, integrating by parts two times, we get

$$\begin{aligned}I &= \lim_{y \rightarrow -\infty} \left[\frac{u}{s} \exp\left(\frac{sZ}{u}\right) \cos(az) \right]_y^0 + \frac{au}{s} \int_{-\infty}^0 \exp\left(\frac{sZ}{u}\right) \sin(az) dz \\ &= \frac{u}{s} + \frac{a^2 u^2}{s^2} \lim_{y \rightarrow -\infty} \left[\exp\left(\frac{sZ}{u}\right) \sin(az) \right]_y^0 - \frac{a^2 u^2}{s^2} I \\ &= \frac{u}{s} - \frac{a^2 u^2}{s^2} I, \\ \Rightarrow (s^2 + a^2 u^2) I &= us, \\ \Rightarrow I &= \frac{us}{s^2 + a^2 u^2}.\end{aligned}$$

So,

$$\xi[\cos(az)] = \frac{1}{u^2} I = \frac{s}{u(s^2 + a^2 u^2)}.$$

Similarly, for $\xi[\sin(az)] = \frac{-a}{(s^2 + a^2 u^2)}$. □

Theorem 4 *The new integral transform of the hyperbolic function such as $\cosh(az)$ and $\sinh(az)$ are*

$$\begin{aligned}\xi[\cosh(az)] &= \frac{s}{u(s^2 - a^2 u^2)}, \\ \xi[\sinh(az)] &= \frac{-a}{(s^2 - a^2 u^2)}.\end{aligned}$$

Lemma 2.2 *(Relation with Laplace transform).*

$$\xi[f(z)](s, u) = \frac{1}{u^2} L[f(-z)]\left(\frac{s}{u}\right),$$

where L denote the Laplace transform.

Proof By using the change of variable $\tau = -z \Rightarrow z = -\tau$, $dz = -d\tau$, when $z \leq 0$ and $\tau \geq 0$, we have

$$\xi[f(z)](s, u) = \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sZ}{u}\right) f(z) dz$$

$$\begin{aligned}
&= \frac{1}{u^2} \int_0^{\infty} \exp\left(-\frac{s\tau}{u}\right) f(-\tau) d\tau \\
&= \frac{1}{u^2} L[f(-\tau)]\left(\frac{s}{u}\right),
\end{aligned}$$

this finishes the proof. $\square$

Definition 2 (Convolution). Let f, g be piecewise continuous functions on $]-\infty; 0]$ of exponential order, then the convolution of functions $(f \star g)$ is defined as:

$$(f \star g)(z) = \int_{-\infty}^z f(\tau) g(z - \tau) d\tau.$$

Theorem 5 (Linearity property and convolution). Let $\alpha f(t)$ and $\beta g(t)$ in the set $\mathcal{F}$, then $(\alpha f(t) + \beta g(t)) \in \mathcal{F}$, where α and β are nonzero real arbitrary constants, and

1- The linearity property is satisfied

$$\xi[\alpha f(t) + \beta g(t)] = \alpha \xi[f(t)] + \beta \xi[g(t)].$$

2- The new integral transform of convolution is satisfied

$$\xi[(f \star g)(z)] = u^2 \xi[f(z)] \cdot \xi[g(z)].$$

Proof 1- It is trivial, and follows by using the definition (2.2).

2- Using the relation between the new transform and the Laplace transform, one has

$$\xi[(f \star g)(z)](s, u) = \frac{1}{u^2} L[(f \star g)(-z)]\left(\frac{s}{u}\right).$$

Next, applying the Laplace transform of convolution

$$L[(f \star g)(z)](s) = L[f(z)](s) \cdot L[g(z)](s).$$

Then, we can obtain

$$\begin{aligned}
\xi[(f \star g)(z)](s, u) &= \frac{1}{u^2} L[(f \star g)(-z)]\left(\frac{s}{u}\right) \\
&= \frac{1}{u^2} L[f(-z)]\left(\frac{s}{u}\right) \cdot L[g(-z)]\left(\frac{s}{u}\right) \\
&= \frac{1}{u^2} u^2 \xi[f(z)](s, u) \cdot u^2 \xi[g(z)](s, u) \\
&= u^2 \xi[f(z)](s, u) \cdot \xi[g(z)](s, u).
\end{aligned}$$

$\square$

Theorem 6 Let a function $f(az)$ be in set $\mathcal{F}$, where $a \in \mathbb{R}$. Then

$$\xi[f(az)] = \frac{1}{a} \Theta\left[\left(\frac{s}{a}, u\right)\right] = \frac{1}{a} \Theta[(s, au)].$$

Proof Using definition (2.2), we have

$$\xi[f(az)] = \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) f(az) dz.$$

We let $az = y$, then $z = \frac{y}{a}$, $adz = dy \Rightarrow dz = \frac{dy}{a}$, substituting in the above expression, yields

$$\xi[f(az)] = \frac{1}{au^2} \int_{-\infty}^0 \exp\left(\frac{sy}{ua}\right) f(y) dy = \frac{1}{a} \Theta\left[\left(\frac{s}{a}, u\right)\right] = \frac{1}{a} \Theta[s, au]. \quad \square$$

Theorem 7 Let a function $f(t)$ be in set $\mathcal{F}$ and its new integral transform is $\xi[f(z)]$, then:

$$\xi[e^{az}f(z)] = \Theta[(s + au; u)], \quad (2.4)$$

where $\xi[f(z)] = \Theta[s, u]$ and a is a real number.

Proof By using the definition (2.2), we have

$$\begin{aligned} \xi[\exp(az)f(z)] &= \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) \exp(az) f(z) dz = \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u} + az\right) f(z) dz \\ &= \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{s + au}{u} z\right) f(z) dz = \Theta[(s + au; u)]. \quad \square \end{aligned}$$

Example 2.1 $\xi[z \exp(az)] = \frac{-1}{s^2} \Big|_{s \rightarrow s+au} = \frac{-1}{(s + au)^2}.$

$$\xi[z^2 \exp(-z)] = \frac{2u}{s^3} \Big|_{s \rightarrow s-u} = \frac{2u}{(s - u)^3}.$$

Theorem 8 (New Integral Transform of Integrals). If

$$B_n(z) = \int_0^{z_1} \int_0^{z_2} \dots \int_0^{z_n} f(z) (dz)^n,$$

where $B_n(z) \in \mathcal{F}$ and $n \in \mathbb{N}$. Then, the new integral transform of $B_n(z)$ is defined as

$$\xi[B_n(z)] = \Theta_n[s, u] = \left(\frac{-u}{s}\right)^n \Theta[s, u].$$

Proof By the principle of mathematical induction, For $n = 1$, we get

$$\xi[B_1(z)] = \Theta_1[s, u] = \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) \left(\int_0^{z_1} f(z) dz\right) dz.$$

Using integration by part, we obtain

$$\xi[B_1(z)] = \Theta_1[s, u] = \left(\frac{-u}{s}\right) \Theta[s, u].$$

Then, for $k \geq n$,

$$\xi[B_{k+1}(z)] = \Theta_{k+1}[(s, u)] = \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) \left(\int_0^{z_1} \int_0^{z_2} \dots \int_0^{z_{k+1}} f(z) (dz)^n \right) dz.$$

Next, using integration by part, we get

$$\xi[B_{k+1}(z)] = \Theta_{k+1}[(s, u)] = \left(\frac{-u}{s}\right) \left(\frac{-u}{s}\right)^k \Theta[(s, u)] = \left(\frac{-u}{s}\right)^{k+1} \Theta[(s, u)].$$

Hence, the proof is finished. $\square$

Theorem 9 (The New Integral Transform of higher order derivatives). Let a function $f(z)$ be in set $\mathcal{F}$ and $\xi[f(z)]$ its new integral transform, then:

$$\begin{aligned} i) \xi[f'(z)] &= \frac{-s}{u} \Theta[(s, u)] + \frac{1}{u^2} f(0); \\ ii) \xi[f^{(n)}(z)] &= (-1)^n \left(\frac{s}{u}\right)^n \Theta[(s, u)] + \frac{(-1)^{n-1}}{u^2} \sum_{k=0}^{n-1} (-1)^k \left(\frac{s}{u}\right)^{n-(k+1)} f^{(k)}(0), n \geq 1. \end{aligned}$$

Proof i) By applying integration by parts, it follows that

$$\begin{aligned} \xi[f'(z)] &= \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) f'(z) dz \\ &= \frac{1}{u^2} \left[\exp\left(\frac{sz}{u}\right) f(z) \right]_{-\infty}^0 - \frac{s}{u^3} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) f(z) dz \\ &= \frac{-s}{u} \Theta[(s, u)] + \frac{1}{u^2} f(0). \end{aligned}$$

ii) By induction technique, for $n = 1$, the property holds true. Let there is an integer j where $j \geq n$. For $j = n$, it is verified that the statement holds true. Now, for $j \geq n$, we proceed by induction to establish the result for $j + 1$ as below:

$$\begin{aligned} \xi[f^{(j+1)}(z)] &= \xi[(f^{(j)})'(z)] \\ &= \frac{-s}{u} \left[(-1)^j \left(\frac{s}{u}\right)^j \Theta[(s, u)] + \frac{(-1)^{j-1}}{u^2} \sum_{k=0}^{j-1} (-1)^k \left(\frac{s}{u}\right)^{j-(k+1)} f^{(k)}(0) \right] \\ &\quad + \frac{1}{u^2} f^{(j)}(0) \\ &= (-1)^{j+1} \left(\frac{s}{u}\right)^{j+1} \Theta[(s, u)] - \frac{(-1)^{j-1}}{u^2} \sum_{k=0}^{j-1} (-1)^k \left(\frac{s}{u}\right)^{j-k} f^{(k)}(0) + \frac{1}{u^2} f^{(j)}(0) \\ &= (-1)^{j+1} \left(\frac{s}{u}\right)^{j+1} \Theta[(s, u)] + \frac{(-1)^j}{u^2} \sum_{k=0}^j (-1)^k \left(\frac{s}{u}\right)^{j-k} f^{(k)}(0). \end{aligned} \quad \square$$

Definition 3 [17] The unit step function or Heaviside function $H(z - a)$ is 0 for $z < a$, has a jump of size 1 at $z = a$ (where we can leave it undefined), and is 1 for $z > a$, in a

formula:

$$H(z-a) = \begin{cases} 0, & \text{for } z < a, \\ 1, & \text{for } z > a, \end{cases}$$

where $a \geq 0$.

Theorem 10 *Let $H(z-a)$ be a Heaviside step function. Then, the new integral transform is defined by*

$$\xi[H(z-a)] = \frac{1}{us} \exp\left(\frac{as}{u}\right). \quad (2.5)$$

Proof By using new integral transform form (2.2), we find

$$\begin{aligned} \xi[H(z-a)] &= \frac{1}{u^2} \int_{-\infty}^a \exp\left(\frac{sz}{u}\right) H(z-a) dz \\ &= \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{sz}{u}\right) H(z-a) dz + \frac{1}{u^2} \int_0^a \exp\left(\frac{sz}{u}\right) H(z-a) dz \\ &= 0 + \frac{1}{u^2} \int_0^a \exp\left(\frac{sz}{u}\right) dz \\ &= \frac{1}{us} \exp\left(\frac{as}{u}\right). \end{aligned} \quad \square$$

Theorem 11 *The new integral transform of partial derivatives is defined by*

$$\begin{aligned} (i) \quad \xi\left[\frac{\partial f(x,z)}{\partial x}\right] &= \frac{d}{dx} \Theta[(x, s; u)] \text{ and } \xi\left[\frac{\partial^n f(x,t)}{\partial x^n}\right] = \frac{d^n}{dx^n} \Theta[(x, s; u)]. \\ (ii) \quad \xi\left[\frac{\partial f(x;z)}{\partial z}\right] &= \frac{-s}{u} \Theta[(s, u)] + \frac{1}{u^2} f(x;0). \\ (iii) \quad \xi\left[\frac{\partial^n f(x;z)}{\partial z^n}\right] &= (-1)^n \left(\frac{s}{u}\right)^n \Theta[(s, u)] + \frac{(-1)^{n-1}}{u^2} \sum_{k=0}^{n-1} (-1)^k \left(\frac{s}{u}\right)^{n-(k+1)} \frac{\partial^k f}{\partial z^k}(x;0), \quad n \geq 1. \end{aligned}$$

Proof (i) By applying Leibniz's rule, we obtain

$$\begin{aligned} \xi\left[\frac{\partial f(x,z)}{\partial x}\right] &= \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{st}{u}\right) \frac{\partial f(x,z)}{\partial x} dt \\ &= \frac{\partial}{\partial x} \left[\frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{st}{u}\right) f(x,z) dz \right] \\ &= \frac{\partial}{\partial x} \Theta[(x, s; u)] = \frac{d}{dx} \Theta[(x, s; u)]. \end{aligned}$$

And

$$\xi\left[\frac{\partial^n f(x,z)}{\partial x^n}\right] = \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{st}{u}\right) \frac{\partial^n f(x,z)}{\partial x^n} dz$$

$$\begin{aligned}
&= \frac{\partial^n}{\partial x^n} \left[\frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{st}{u}\right) f(x, z) dz \right] \\
&= \frac{\partial^n}{\partial x^n} \Theta[(x, s; u)] = \frac{d^n}{dx^n} \Theta[(x, s; u)].
\end{aligned}$$

(ii) Applying integration by parts, we have

$$\begin{aligned}
\xi \left[\frac{\partial f(x, z)}{\partial z} \right] &= \frac{1}{u^2} \int_{-\infty}^0 \exp\left(\frac{st}{u}\right) \frac{\partial f(x, z)}{\partial z} dz \\
&= \frac{1}{u^2} \left[\exp\left(\frac{st}{u}\right) f(x; z) \right]_{-\infty}^0 - \frac{s}{u^3} \int_{-\infty}^0 \exp\left(\frac{st}{u}\right) f(x; z) dz \\
&= \frac{-s}{u} \Theta[(s, u)] + \frac{1}{u^2} f(x; 0).
\end{aligned}$$

(iii) We use mathematical induction in the proof. So, for $n = 1$, the property is satisfied as in part (ii).

Suppose there exists an integer j , such that $j \geq n$, for $j = n$, it is verified as below:

$$\xi \left[\frac{\partial^j f(x; z)}{\partial z^j} \right] = (-1)^j \left(\frac{s}{u} \right)^j \Theta[(s, u)] + \frac{(-1)^{j-1}}{u^2} \sum_{k=0}^{j-1} (-1)^k \left(\frac{s}{u} \right)^{j-(k+1)} \frac{\partial^k f}{\partial z^k}(x; 0), j \geq 1.$$

Then, for $j \geq n$, we have

$$\begin{aligned}
&\xi \left[\frac{\partial^{(j+1)} f(x, z)}{\partial z^{(j+1)}} \right] = \xi \left[\frac{\partial}{\partial z} \left(\frac{\partial^j f(x, z)}{\partial z^j} \right) \right] \\
&= \frac{-s}{u} \xi \left[\frac{\partial^j f}{\partial z^j}(x; z) \right] + \frac{1}{u^2} \frac{\partial^j f}{\partial z^j}(x; 0) \\
&= \frac{-s}{u} \left[(-1)^j \left(\frac{s}{u} \right)^j \Theta[(s, u)] + \frac{(-1)^{j-1}}{u^2} \sum_{k=0}^{j-1} (-1)^k \left(\frac{s}{u} \right)^{j-(k+1)} \frac{\partial^k f}{\partial z^k}(x; 0) \right] + \frac{1}{u^2} \frac{\partial^j f}{\partial z^j}(x; 0) \\
&= (-1)^{j+1} \left(\frac{s}{u} \right)^{j+1} \Theta[(s, u)] + \frac{(-1)^{(j+1)-1}}{u^2} \sum_{k=0}^{j-1} (-1)^k \left(\frac{s}{u} \right)^{(j+1)-(k+1)} \frac{\partial^k f}{\partial z^k}(x; 0) + \frac{1}{u^2} \frac{\partial^j f}{\partial z^j}(x; 0), \\
&= (-1)^{j+1} \left(\frac{s}{u} \right)^{j+1} \Theta[(s, u)] + \frac{(-1)^{(j+1)-1}}{u^2} \sum_{k=0}^{(j+1)-1} (-1)^k \left(\frac{s}{u} \right)^{j-k} \frac{\partial^k f}{\partial z^k}(x; 0).
\end{aligned}$$

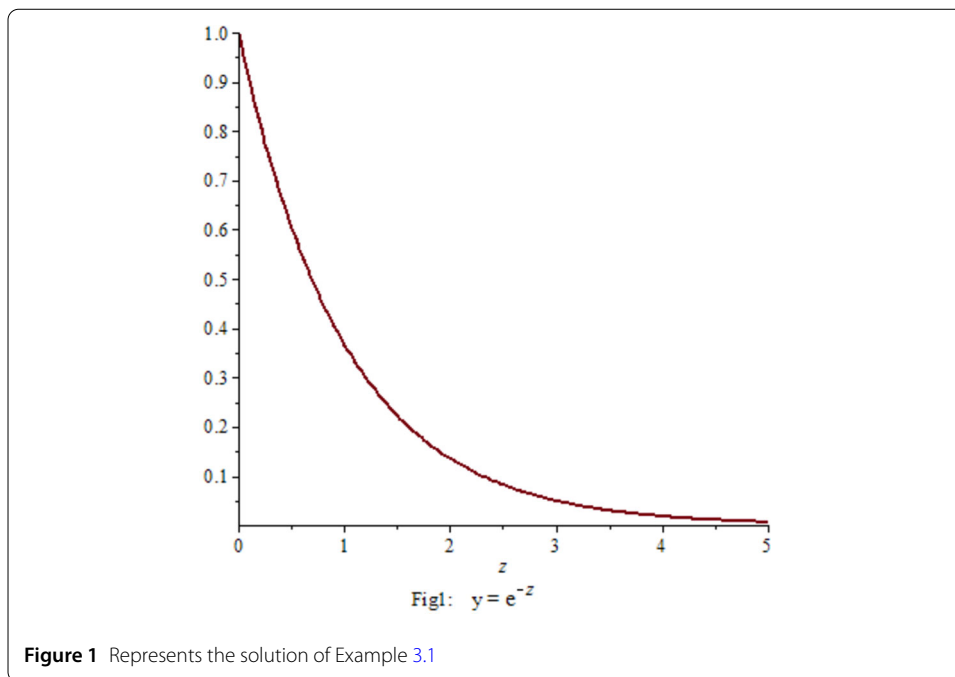
Hence, by mathematical induction it is true for each $n \geq 1$. $\square$

3 Some application of new integral transform

This section explores the application of the new integral transform in solving differential equations. The following straightforward examples demonstrate that:

Example 3.1 Examine the following first-order ordinary differential equation:

$$\frac{df(z)}{dz} + f(z) = 0, \tag{3.1}$$



with initial condition

$$f(0) = 1. \quad (3.2)$$

Applying the new integral transform to Eq. (3.1), we obtain

$$\xi \left[\frac{df(z)}{dz} \right] + \xi [f(z)] = 0.$$

Using Eq. (3.2), yields

$$\frac{-s}{u} \Theta[(s, u)] + \frac{1}{u^2} + \Theta[(s, u)] = 0.$$

Then,

$$\begin{aligned} \left(\frac{-s}{u} + 1 \right) \Theta[(s, u)] &= \frac{-1}{u^2}, \\ \Rightarrow \Theta[(s, u)] &= \frac{1}{u(s-u)}. \end{aligned} \quad (3.3)$$

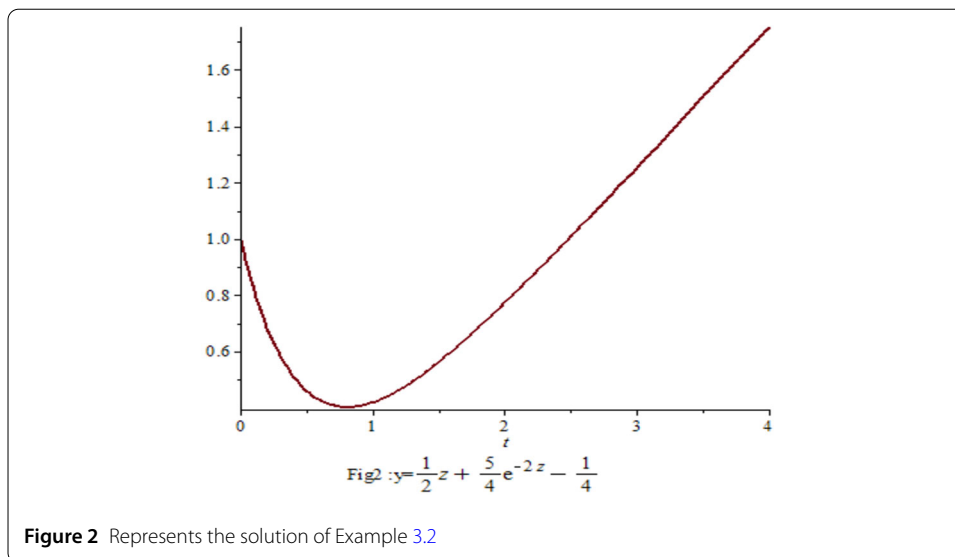
By taking the inverse of new integral transform of Eq. (3.3), we obtain

$$f(z) = \exp(-z),$$

which plots in Fig. 1.

Example 3.2 Consider the following differential equation

$$\frac{dy}{dz} + 2y = z, \quad y(0) = 1. \quad (3.4)$$



Applying the new integral transform on Eq. (3.4), we find

$$\xi\left[\frac{dy}{dz}\right] + 2\xi[y] = \xi[z].$$

So, we have

$$\frac{-s}{u}\Theta[(s, u)] + \frac{1}{u^2} + 2\Theta[(s, u)] = \frac{-1}{s^2}.$$

After simplification, we obtain

$$\Theta[(s, u)] = \frac{u^2 + s^2}{us^2(s - 2u)} = \frac{-1}{2s^2} + \frac{5}{4} \frac{1}{u(s - 2u)} - \frac{1}{4su}. \quad (3.5)$$

The inverse of new integral transform ξ^{-1} of equation (3.5) gives the solution

$$y(z) = \frac{1}{2}z + \frac{5}{4}\exp(-2z) - \frac{1}{4},$$

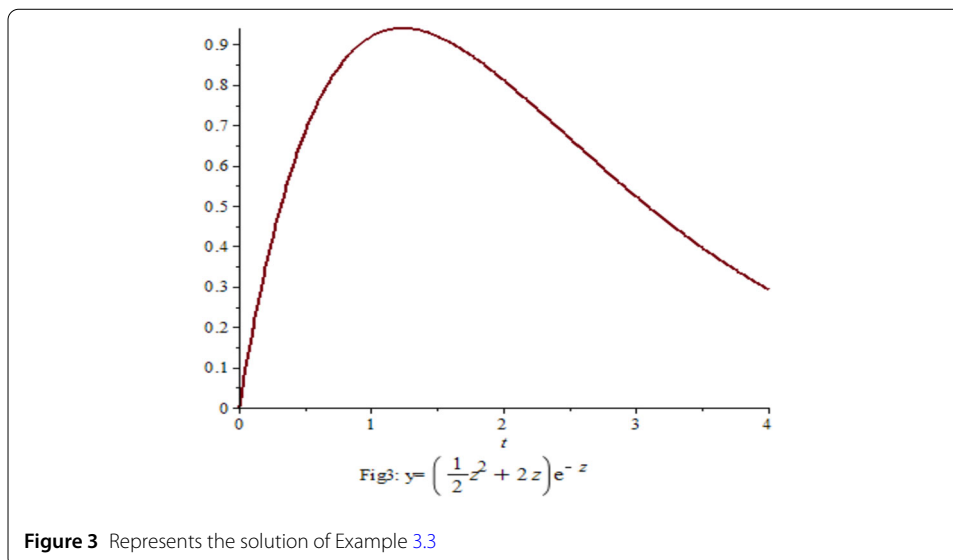
which is plotted in Fig. 2.

Example 3.3 Examine the following second-order non-homogeneous ordinary differential equation:

$$y'' + 2y' + y = \exp(-z), \quad (3.6)$$

with the initial conditions

$$y(0) = 0, \quad y'(0) = 2. \quad (3.7)$$



By applying the new integral transform of Eq. (3.6), we get

$$\begin{aligned} \xi[y''(z)] + 2\xi[y'(z)] + \xi[y(z)] &= \xi[\exp(-z)]. \\ \Leftrightarrow \frac{s^2}{u^2} \Theta[(s, u)] + \frac{1}{u^2} y'(0) - \frac{s}{u^3} y(0) - \frac{2s}{u} \Theta[(s, u)] + \frac{2}{u^2} y(0) + \Theta[(s, u)] &= \frac{1}{u(s-u)}, \end{aligned}$$

After simplification, the result is

$$\Theta[(s, u)] = \frac{u}{(s-u)^3} - \frac{2}{(s-u)^2}. \quad (3.8)$$

By taking ξ^{-1} of both sides of Eq. (3.8), we obtain the general solution

$$\begin{aligned} y(z) &= \xi^{-1} \left[\frac{u}{(s-u)^3} \right] + 2\xi^{-1} \left[\frac{-1}{(s-u)^2} \right], \\ &= \frac{1}{2} z^2 \exp(-z) + 2z \exp(-z) = \left(\frac{1}{2} z^2 + 2z \right) \exp(-z), \end{aligned}$$

which is plotted in Fig. 3.

Example 3.4 Examine the following second-order non-homogeneous

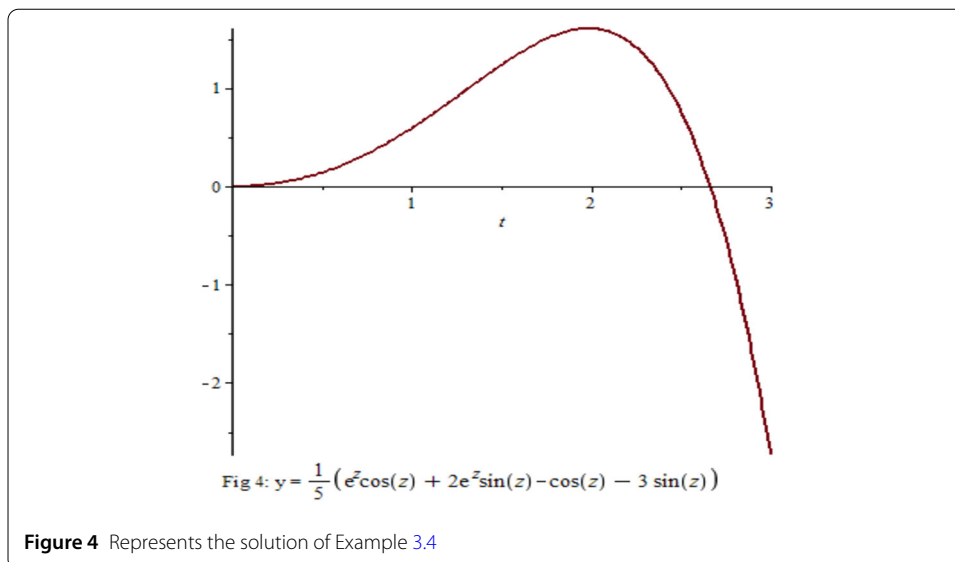
$$y'' + y = \exp(z) \cos z, \quad (3.9)$$

with initial conditions

$$y(0) = 0, \quad y'(0) = 0, \quad (3.10)$$

We apply the new integral transform of Eq. (3.9), we obtain

$$\begin{aligned} \xi[y''(z)] + \xi[y(z)] &= \xi[\exp(z) \cos z]. \\ \Leftrightarrow \frac{s^2}{u^2} \Theta[(s, u)] + \frac{1}{u^2} y'(0) - \frac{s}{u^3} y(0) + \Theta[(s, u)] &= \frac{s+u}{u((s+u)^2 + u^2)}. \end{aligned}$$



After simplifying, we have

$$\Theta[(s, u)] = \frac{u(s+u)}{(s^2+u^2)((s+u)^2+u^2)}.$$

By decomposition, we get

$$\Theta[(s, u)] = \frac{1}{5} \left[\frac{(s+u)}{u((s+u)^2+u^2)} - \frac{2}{((s+u)^2+u^2)} - \frac{s}{u(s^2+u^2)} + \frac{3}{(s^2+u^2)} \right]. \quad (3.11)$$

By taking ξ^{-1} of both sides of Eq. (3.11), we obtain the solution

$$y(z) = \frac{1}{5} [\exp(z) \cos z + 2 \exp(z) \sin z - \cos z - 3 \sin z], \quad (3.12)$$

which is plotted in Fig. 4.

Example 3.5 Consider the following standard diffusion equation

$$\frac{\partial^2 u(x; z)}{\partial x^2} + \frac{\partial u(x; z)}{\partial z} = 0, \quad x \in \mathbb{R}, z > 0, \quad (3.13)$$

with $u(x, z)$ as a field variable, supported by the initial conditions

$$u(0; z) = u(1; z) \rightarrow 0, z \rightarrow +\infty, u(x; 0^+) = 2e^{-x}. \quad (3.14)$$

We apply the new integral transform of Eq. (3.13), we get

$$\frac{d^2}{dx^2} \Theta[(x; s; u)] - \frac{s}{u} \Theta[(x; s; u)] + \frac{1}{u^2} u(x; 0) = 0. \quad (3.15)$$

According to the initial conditions, we obtain

$$\frac{d^2}{dx^2} \Theta[(x; s; u)] - \frac{s}{u} \Theta[(x; s; u)] = -\frac{2}{u^2} e^{-x}, \quad (3.16)$$

Then, the general solution of Eq. (3.16) is

$$\Theta_H[(x; s; u)] = \Theta_H[(x; s; u)] + \Theta_P[(x; s; u)],$$

where $\Theta_H[(x; s; u)]$ is the solution the homogeneous equation

$$\frac{d^2}{dx^2} \Theta[(x; s; u)] - \frac{s}{u} \Theta[(x; s; u)] = 0,$$

which is as

$$\Theta_H[(x, s; u)] = c_1 e^{\sqrt{\frac{s}{u}}x} + c_2 e^{-\sqrt{\frac{s}{u}}x}. \quad (3.17)$$

Applying the boundary conditions on Eq. (3.15), we obtain

$$c_1 = 0, \quad c_2 = 0.$$

So, $\Theta_H[(x, s; u)] = 0$.

And the particular solution $\Theta_P[(x, s; u)]$ is

$$\Theta_P[(x, s; u)] = ae^{-x}, \quad a \in \mathbb{R}. \quad (3.18)$$

By substituting in Eq. (3.18), we get

$$a = \frac{-2}{u(u-s)}. \quad (3.19)$$

So,

$$\Theta[(x, s; u)] = \Theta_H[(x, s; u)] + \Theta_P[(x, s; u)] = \frac{-2}{u(u-s)} e^{-x}, \quad (3.20)$$

By taking ξ^{-1} of both sides of Eq. (3.20), we find the solution

$$u(x, z) = 2e^{-z-x}$$

which is plotted in Fig. 5.

In what follows, we'd like to present an example where the Laplace transform cannot directly solve an integral equation, but it can be solved using the proposed transformation.

Example 3.6 Consider the following Volterra integral equation:

$$f(z) - \lambda \int_{-\infty}^z e^{a(z-\tau)} f(\tau) d\tau = e^{bz}, \quad (3.21)$$

for $f(z)$ defined on $z \leq 0$, and $a - b - \lambda \neq 0$, $a - b \neq 0$, $\lambda \neq 0$.

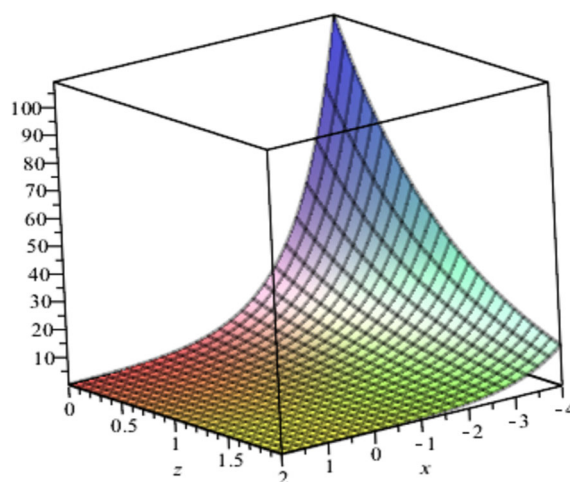
Fig5: $y = 2e^{-z-x}$

Figure 5 Represents the solution of Example 3.5

Notice that the equation is a Volterra-type equation of the second kind, but defined on $]-\infty, 0]$. So, the Laplace transform cannot be used directly, because the integral equation is defined out of the Laplace transform domain, namely from $-\infty$ to z (not from 0 to z).

Since the kernel is given by $k(z) = e^{az}$, $z \leq 0$.

Computing $\xi[k(z)]$, one has

$$\xi[k(z)](s, u) = \frac{1}{u(s + au)},$$

Also, the integral $\int_{-\infty}^z f(\tau)k(z-\tau)d\tau$, is a convolution defined on $]-\infty, 0]$.

Thus,

$$\xi[(k \star f)(z)](s, u) = u^2 \frac{1}{u(s + au)} \xi[f(z)],$$

and the new integral transform of right-hand side of (3.21) is:

$$\xi[e^{bz}] = \frac{1}{u(s + bu)}.$$

Therefore, the new integral transform of equation (3.21) is

$$\xi[f(z)] - \lambda \frac{u}{(s + au)} \xi[f(z)] = \frac{1}{u(s + bu)},$$

so,

$$\xi[f(z)] = \frac{(s + au)}{u(s + bu)(s + au - \lambda u)}. \quad (3.22)$$

Let us consider

$$\frac{(s + au)}{u(s + bu)(s + au - \lambda u)} = \frac{A}{u(s + bu)} + \frac{B}{u(s + au - \lambda u)},$$

where $A = \frac{a-b}{a-b-\lambda}$, and $B = \frac{\lambda}{b-a+\lambda}$.

Applying the inverse transform for (3.22), we get

$$f(z) = Ae^{bz} + Be^{(a-\lambda)z}, \quad z \leq 0.$$

4 Results and discussion

The new integral transform allows us to solve ordinary differential equations (ODEs) and integro-differential equations defined on $z \leq 0$ in a way that the classical Laplace transform cannot handle directly. Here are the key results and why they matter:

1. Many physical problems (e.g., pre-history, pre-stressed systems, backward time models) require $z \leq 0$. The new transform is defined on $]-\infty, 0]$, this is the main theoretical contribution (it opens the door to direct solving ODEs on the negative half-axis).
2. Differentiation property (Main tool for ODEs), which transforms a differential equation into an algebraic equation.
3. The results of higher order derivatives allow us to solve the n th order linear ODEs systematically.
4. Handles boundary conditions at $z = 0^-$, because the transform uses $f(0)$ and $f^{(k)}(0)$, initial conditions specified at $z = 0^-$ are naturally included - no need to flip time axis artificially.

5 Conclusion

In this paper, the authors explored the use of a new transform to treat ordinary and partial differential equations. The study presents new formulas of this transform for elementary functions, examines their key properties, and applies them to various problems. The main contribution of this transform can be summarized as follows:

- This new transform generalizes the Laplace transform method to the domain $(-\infty, 0]$, making it possible to solve ordinary differential equations, integro-differential equations, and Volterra equations that are otherwise difficult to directly handle with traditional transforms.
- It preserves all essential operational properties, such as linearity, shifting, differentiation, and convolution, also adapts them to negative time, automatically incorporating initial conditions at $z = 0^-$.
- As a result, differential equations on the negative half-axis are converted into simple algebraic equations in the s -domain, allowing direct, systematic solutions.
- This makes the new transform particularly valuable for modeling systems with delayed effects, backward-time dynamics, and boundary conditions specified before $z = 0^-$, something that the classical Laplace transform can not do without cumbersome time-reversal techniques.

The results indicate that this new transform constitutes an efficient integral transformation for solving such equations. Moreover, all solutions derived from the specified problems satisfy their respective equations after substitution. In the future, this new transform offers potential for broader applications in solving differential equations.

Author contributions

Every author contributed equally to each part of the paper.

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Data availability

No datasets were generated or analysed during the current study.

Declarations**Ethics approval and consent to participate**

Not applicable.

Competing interests

The authors declare no competing interests.

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